

Probabilistic Robotics

Exercises

16/10/2024

3 Exercise: Watch my robot

We have a unicycle robot that is controlled in translational and rotational displacements $\Delta\rho$ and $\Delta\theta$. The displacements (so the velocities) vary mildly, but we do not have access to them. The robot is observed by N stations located at known position. Each station senses the “bearing of the robot”

We want to track the robot with an EKF (determine its position).

4 Exercise: 2D GPS with global clock

We have a unicycle robot that is controlled in translational and rotational displacements $\Delta\rho$ and $\Delta\theta$, that we can access. The robot is equipped with an antenna capable of receiving the signal emitted by a set of beacons, that move on the plane. In the message, each beacon encodes:

- unique ID
- its position on the plane $\mathbf{p}_i$
- the time t_i^s at which the beacon transmits the signal, on a global clock.

The robot might not sense all the beacons at all times.

We want to track the robot with an EKF (determine its position), under the assumption that it has access to the global clock. In this case, we can assume that, when the robot receives a message it immediately timestamps it to get the arrival time t_i^a . We remark that knowing the location of the beacon $\mathbf{p}_i$, the sending time t_i^s , and the robot location $\mathbf{x}$, we can compute the expected arrival time.